

Quick Check

Lesson 4: Proteins and Lipids: Do They Cause the Glucose Spike?

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.2: Chemicals of Life | Lesson 4

Before the tests, a learner says: "Iodine shows starch in ugali, so I will use iodine to find out whether beans contain protein." Which response is correct?

- A** Iodine will turn violet if protein is present, so the plan works
- B** Iodine tests for all food chemicals, so it will show protein and fat as well
- C** Iodine tests for starch only, so a different test (the biuret test) is needed for protein
- D** Protein cannot be tested for in the laboratory, so beans must be judged by taste

A group adds 2 cm³ of dilute sodium hydroxide and then 2-3 drops of dilute copper sulphate to a tube of bean paste mixed with water. The mixture turns violet (purple). What does this result show?

- A The beans contain starch
- B The beans contain protein
- C The beans contain lipids (fats and oils)
- D The beans contain reducing sugar

A learner rubs crushed groundnuts on a piece of brown paper, dries it gently and holds it up to the light. A translucent (see-through) spot is left on the paper. What does this show?

- A** The groundnuts contain starch
- B** The groundnuts contain protein
- C** The groundnuts contain lipids (oil)
- D** The groundnuts contain only water, which has made the paper thin

Which row correctly matches each food chemical to the smaller units it is built from?

- A** Carbohydrates: glucose units; proteins: amino acids; lipids: fatty acids and glycerol
- B** Carbohydrates: glucose units; proteins: fatty acids and glycerol; lipids: amino acids
- C** Carbohydrates: amino acids; proteins: glucose units; lipids: fatty acids and glycerol
- D** Carbohydrates: fatty acids and glycerol; proteins: glucose units; lipids: amino acids

Wanjiru's ugali contains about 70 g of carbohydrate and about 10 g of fat. About how many times more carbohydrate than fat is this?

A 60 times

B 80 times

C 7 times

D 0.14 times

Beans, milk and groundnuts are not the main cause of the glucose spike after a meal. Why do they still matter for the body?

- A** They are turned into glucose immediately, so they cause the spike as well
- B** They only add sweetness, so they have no use in the body
- C** They replace carbohydrates completely, so ugali is not needed
- D** Proteins in them are used for growth and repair of body tissues, and lipids are used for energy storage

Learners are updating their flow-diagram model of what happens to a meal of ugali, beans and cooking oil. Which arrow should be drawn thick, as the main cause of the glucose spike in the blood?

- A Proteins (beans, nyama choma) to glucose in blood
- B Carbohydrates (starch in ugali) to glucose in blood
- C Proteins and lipids to glucose in blood, equally with carbohydrates
- D Lipids (cooking oil, groundnuts) to glucose in blood

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).